



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105



Student Name: B. Santhosh Kumar

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Course Code: CSA1709

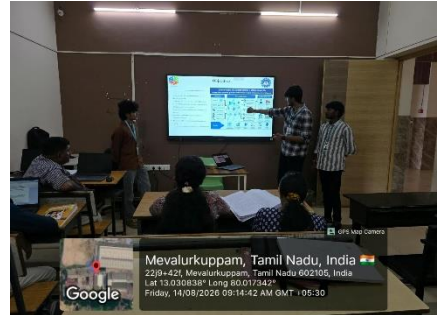
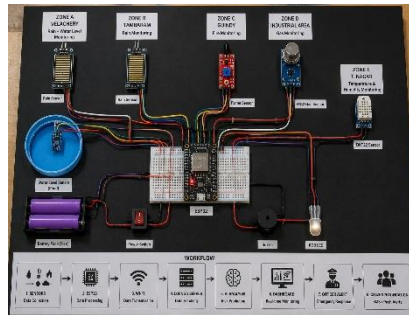
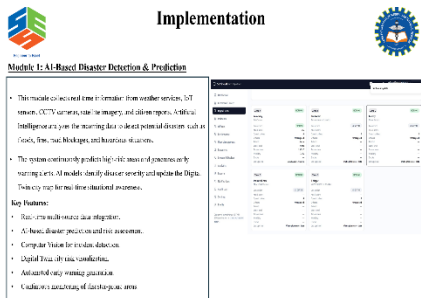
Slot: B

Course Name: Artificial Intelligence

Course Faculty: Dr P Rajaram

Project Title: SentinelAI: A Multi-Agent Urban Digital Twin for Predictive Disaster Intelligence and Autonomous Emergency Response

Module Photographs:



Project Description:

SentinelAI is an AI-powered smart-city platform for real-time disaster monitoring, prediction, and emergency response. ESP32-based IoT sensors collect temperature, humidity, rainfall, water-level, flame, and gas data, which are transmitted to a Django backend, validated, processed, and stored in PostgreSQL. The system evaluates zone-specific risks using configurable thresholds, generates alerts and incidents, and supports officer and resource allocation. A Digital Twin dashboard provides centralized monitoring of zones, live sensor data, device health, risk levels, and incidents. Citizen reporting, notifications, analytics, and an Officer Incentive Points Dashboard further improve coordination, accountability, and emergency response efficiency.

Student Signature

Guide Signature